

Simulating a Diesel Engine

IN ANSYS FORTE

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I would like to use this opportunity to extend my heartfelt appreciation to the Department of Mechanical Engineering at the Indian Institute of Technology Jodhpur for supplying the necessary facilities and resources. Because of their aid, the implementation of complex actions essential to accomplishing the project's objectives was completed more easily.

Introduction

Contemporary investigations within the engineering domain are presently addressing the intricate challenges associated with dual-fuel combustion in internal combustion engines. This approach seeks to amalgamate the efficiency characteristics akin to diesel combustion with gasoline as the primary fuel source. The envisaged benefits include a potential 50% efficiency enhancement coupled with concomitant emissions reductions. Conventional dual-fuel combustion configurations entail a judicious pilot injection of a readily ignitable liquid fuel, typically diesel, into a lean air mixture containing a more volatile fuel, such as gasoline, which is port-injected to ensure optimal pre-ignition air-fuel mixing. The computational tool Ansys Forte is instrumental in this pursuit, employing accurate fuel models to proficiently simulate the intricate interplay of ignition and flame propagation phenomena.

In spark-ignited engines, flame propagation is conventionally instigated by a spark event. However, in dual-fuel scenarios, the initiation of flame propagation is facilitated by the injection and auto-ignition of the liquid pilot fuel. Ansys Forte's simulation framework accommodates the concurrent consideration of auto-ignition and flame-propagation modes. The efficacy of this approach is underscored by the software's ability to leverage precise fuel models in predicting ignition under diverse temperature and pressure conditions. The flame-propagation model further integrates locally computed turbulent flame speed derived from fuel-specific calculations utilising comprehensive kinetics models.

The Ansys Forte simulation workflow offers a versatile choice between a body-fitted mesh and an automatically generated mesh. The present tutorial emphasises the utilization of a mesh generated via the Sector Mesh Generator. This methodology strategically exploits the periodicity and symmetry inherent in the injector nozzle-hole pattern and cylinder, permitting the representation of the complete geometry through a reduced sector (e.g., a 60° sector for a six-hole injector). This pragmatic approach significantly diminishes the computational burden by reducing mesh size, thereby enhancing the efficiency of simulations relative to a comprehensive 360° mesh configuration.

- **Objective: -**

- The Primary Objective is to compare the output parameters of a diesel engine running Pure Diesel/Bio-diesel/50% Biodiesel/20% Biodiesel.
- Ansys Forte was selected for simulating a diesel engine as it had all the necessary simulation settings.
- This simulation aims to study the effect of different fuels on a diesel engine.

METHODOLOGY

- **Data Collection: -**

Initially, I initiated the research process by immersing myself in ANSYS Forte, undertaking a comprehensive exploration through tutorials and research papers. This served as a foundational step to familiarize myself with the intricacies of the software.

- **Tutorial Implementation: -**

Subsequently, I engaged in hands-on learning by working on tutorial files, a practical exercise that significantly contributed to my understanding of fundamental aspects within the ANSYS Forte framework.

- **Geometry Development: -**

To implement the acquired knowledge, I proceeded to generate the geometry file, drawing inspiration from pertinent tutorials to ensure a comprehensive understanding of the geometric representation.

- **Mesh Generation: -**

In the meshing phase, a meticulous approach was adopted involving the utilization of five distinct mesh types for diverse geometry components. Surface refinement, line refinement, cone refinement, point refinement, and solution-adaptive meshing techniques were employed to optimize the mesh quality.

- **Simulation Setup: -**

The chemical composition for the simulation was derived from the default Ansys library, specifically the "Gasoline-ethanol_4comp_179sp__knocking__soot-pseudo-gas.cks" set. The transport model was maintained at default settings. The spray model incorporated a vaporization model, featuring a solid cone injector equipped with a nozzle facilitating two injections. Custom values were assigned to the velocity profile of the injection.

- **Boundary Conditions: -**

Four boundary conditions were specified, including one periodic boundary and three distinct wall boundaries (linear, head, and piston). These conditions were carefully configured to accurately represent the physical constraints within the simulation domain.

- **Initial Conditions: -**

Customization of the fuel composition was undertaken, involving the assignment of mole fractions to specific species. Adjustments were made to temperature, pressure, velocity, and other pertinent parameters to align with the desired simulation objectives.

- **Simulation Controls: -**

Critical simulation controls were established, delineating the simulation endpoints and incorporating features such as auto-induced flame propagation and dynamic cell clustering. Output frequency was set at intervals of 5 degrees to capture detailed information throughout the simulation. Additionally, specific species were selected for spatially resolved species analysis, complemented by other relevant configurations to enhance the simulation's comprehensiveness.

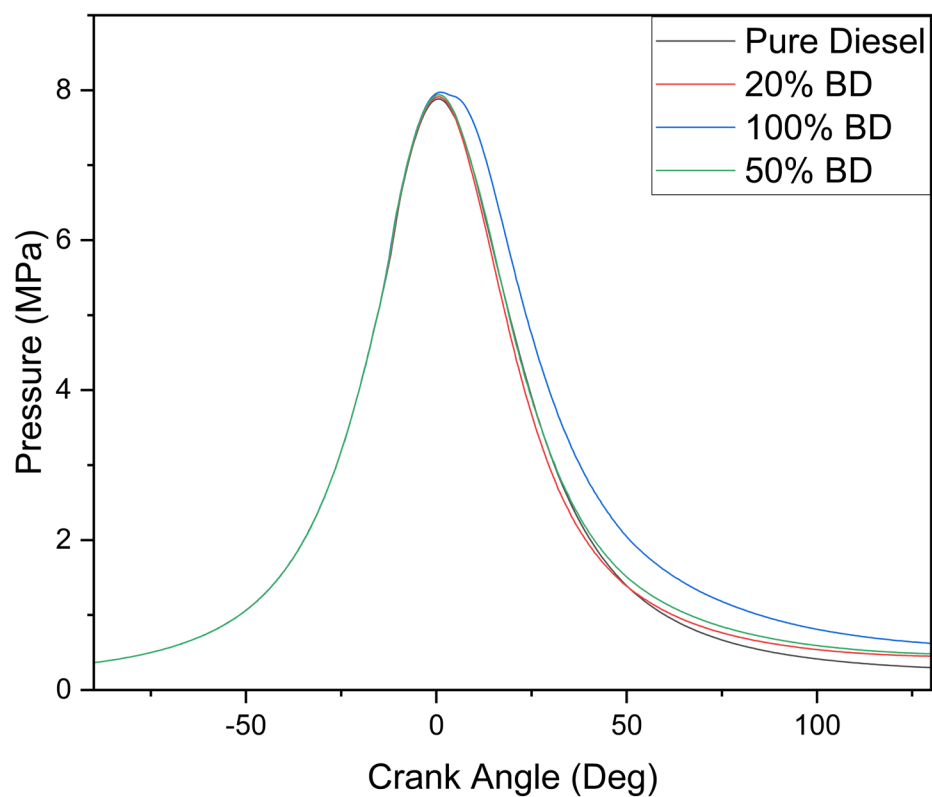
- **Biodiesel Perpetration: -**

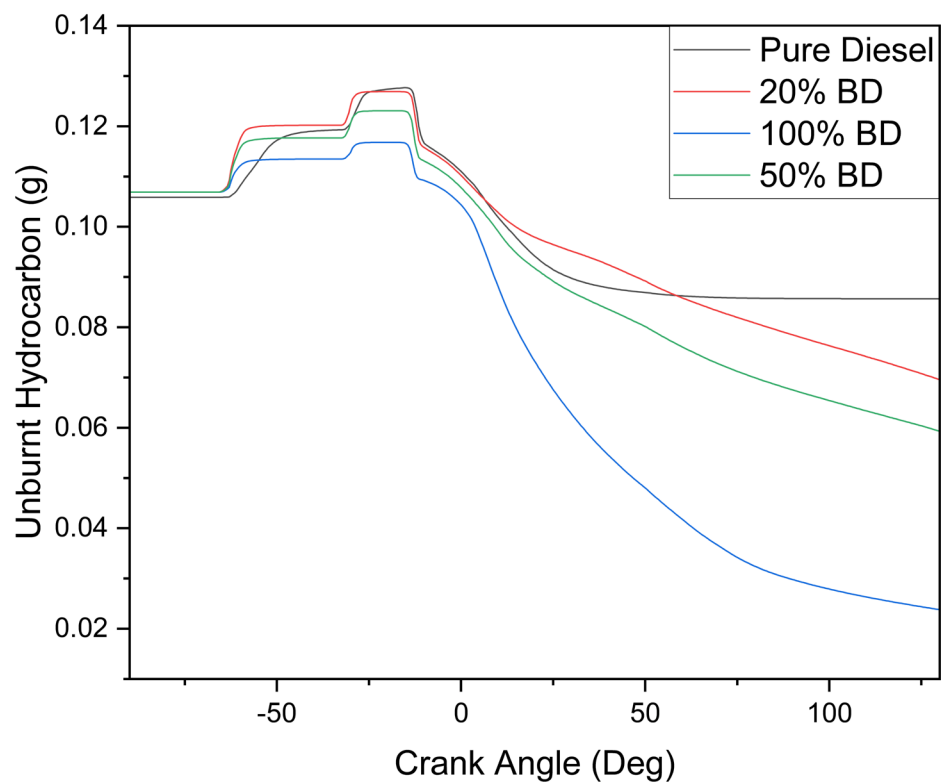
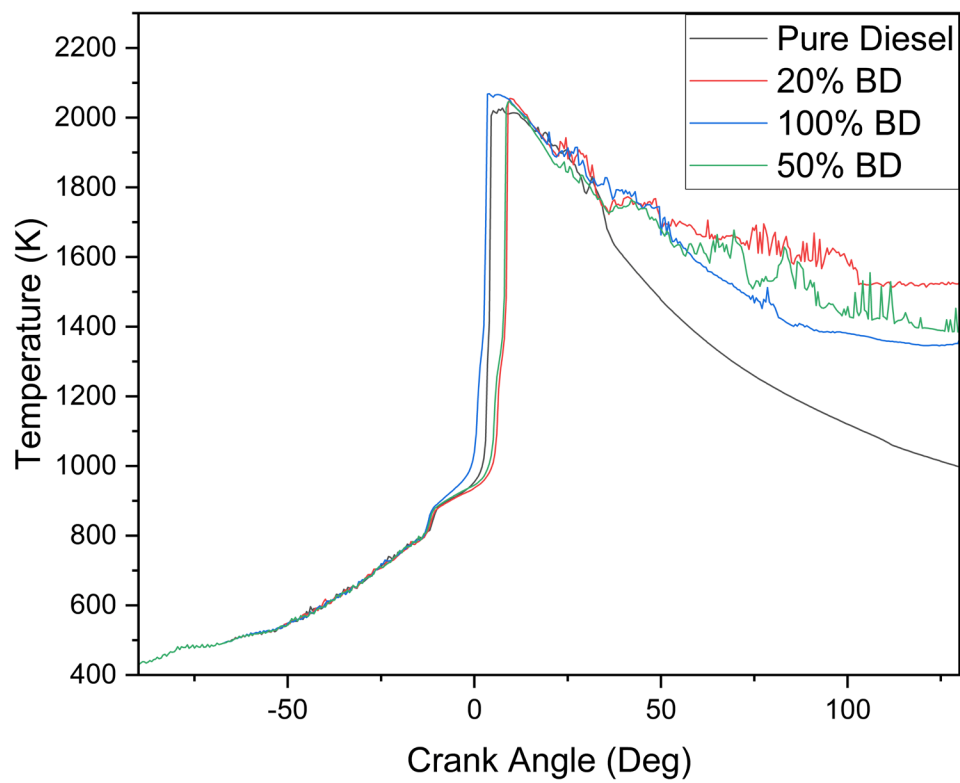
- A non-Ansys mechanism was chosen for simulating fuels due to specific requirements or preferences of the study or project.
- The acquired mechanism files, which include input, thermos, and transport data, were converted to formats compatible with ANSYS software using Cheakmin-Pro or a similar tool. This conversion ensures seamless integration of the mechanism into the ANSYS environment for further analysis and simulation.
- A pure biodiesel composition of 50% n-heptane, 25% methyl palmitate (MD), and 25% methyl oleate (MD9D) by volume was defined. This composition serves as the simulation's basis and represents a common biodiesel blend.

- The physical properties of the individual components, namely n-heptane, methyl palmitate (MD), and methyl oleate (MD9D), were assigned to their respective chemical representations (e.g., C₇H₁₆ for n-heptane) within the ANSYS environment.
- The volume fractions of the fuel components were converted to mass fractions to facilitate the simulation process. This conversion allows for a consistent representation of the fuel mixture based on mass rather than volume, which is often more relevant for combustion simulations.
 - MD9D
 - Volume = 25% = 0.25cm²
 - Density = 0.873 g/cm³
 - Mass = 0.873 x 0.25 = 0.28
 - MD
 - Volume = 25% = 0.25cm²
 - Density = 0.875 g/cm³
 - Density = 0.875 x 0.25 = 0.28
 - C₇H₁₆
 - Volume = 50% = 0.50cm²
 - Density = 0.684 g/cm³
 - Mass = 0.684 x 0.50 = 0.44
- The mass fractions of each component (C₇H₁₆, MD, and MD9D) were calculated based on their respective volume fractions. the volume fractions were 0.50, 0.25, and 0.25 for n-heptane, MD, and MD9D, respectively; the corresponding mass fractions might be calculated as 0.44, 0.28, and 0.28, respectively.
- All simulations were performed using a custom mechanism, which incorporates the specific chemical kinetics and thermodynamics required for modelling the combustion behaviour of the chosen fuel compositions.

Results

The simulation kept running for quite some time; the pressure and the rate of heat released were mainly analysed, with some other factors; the graphs are given below.





Resources Used

- [Balaji et al](#)
- [Som et al](#)
- [Gan et al](#)
- [Pepiot et al](#)
- [Yang et al](#)
- [Ansys Forte](#)